

The empirical recurrence for OEIS A297510: recovering a conjecture that is not written down

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Abstract

OEIS A297510 records “Empirical recurrence of order 59”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 121$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297510 is “Number of $5 \times n$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 1 or 2 neighboring 1s.”. It has offset 1 and begins

1, 280, 2534, 14201, 270845, 3624914, 35049871, 431068698, ...

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Dec 31 2017 07:34:54, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 121$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 121$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 242$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	10	c_{16}	−2286449979	c_{31}	−970456252641	c_{46}	5334531006
c_2	−11	c_{17}	2567960855	c_{32}	−1497909244460	c_{47}	4933049500
c_3	675	c_{18}	−1326686900	c_{33}	831028205921	c_{48}	−3016035914
c_4	−2373	c_{19}	−4640037118	c_{34}	706271486776	c_{49}	−972434046
c_5	−7532	c_{20}	1622392936	c_{35}	2722884653546	c_{50}	−2170574167
c_6	−48225	c_{21}	−68824994187	c_{36}	−165696659281	c_{51}	−1208984889
c_7	134847	c_{22}	140052456013	c_{37}	439896396353	c_{52}	117380707
c_8	329795	c_{23}	−39663860745	c_{38}	−1692212524120	c_{53}	50628701
c_9	4372582	c_{24}	156889381325	c_{39}	−429248037763	c_{54}	58880609
c_{10}	−10376509	c_{25}	−426122618325	c_{40}	−522065809017	c_{55}	7811214
c_{11}	−1124188	c_{26}	−102267184803	c_{41}	470668052525	c_{56}	−330576
c_{12}	−96135888	c_{27}	493271071079	c_{42}	117772002063	c_{57}	33244
c_{13}	154853204	c_{28}	795315988800	c_{43}	198127344463	c_{58}	15036
c_{14}	−45939875	c_{29}	561716733756	c_{44}	−21750297955	c_{59}	−120
c_{15}	1270767149	c_{30}	−1668813686887	c_{45}	2288444685		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59 + 4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A297510.
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